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A Common Representative Intermediates (CRI) mechanism for VOC degradation. Part 1: Gas phase mechanism development

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ABSTRACT

A reduced mechanism describing ozone formation from the tropospheric degradation of methane and 115 emitted non-methane hydrocarbons and oxygenated volatile organic compounds has been developed, using the Master Chemical Mechanism version 3.1 (MCM v3.1) as a reference benchmark. The Common Representative Intermediates mechanism version 2 (CRI v2) has been built up on a compound-by-compound basis, with the performance of its chemistry optimised for each compound in turn by comparison with that of MCM v3.1, using a series of five-day box model simulations. The resultant mechanism contains 1183 reactions of 434 chemical species, i.e., ca. 10% of the number of reactions and species to degrade the same set of VOCs in MCM v3.1. Similarly to CRI v1 (Jenkin et al., 2002a), a key assumption in the CRI v2 construction methodology is that the potential for ozone formation from a given volatile organic compound (VOC) is related to the number of reactive (i.e., C—C and C—H) bonds it contains. This index allows a series of generic intermediates to be defined, with each being used as a "common representative" for a large set of species possessing the same index, as formed in detailed mechanisms such as the MCM. The performance of CRI v2 is shown to compare well with that of MCM v3.1 for a wide range of ambient conditions, which consider variations in VOC/NO_x emissions ratio over ranges of 32 and 400 for anthropogenic and biogenic VOCs, respectively, in box model simulations; and in simulations of the TORCH 2003 campaign in the southern UK using a photochemical trajectory model, which considers notable ranges in the relative and absolute emissions of NO_x and VOCs, and in the relative contributions of anthropogenic and biogenic species to the VOC total. CRI v2 is a reduced mechanism of intermediate complexity, which is traceable to MCM v3.1, and which provides the basis for further systematic reduction. Two companion papers consider further reduction of CRI v2 through emissions lumping and development and assessment of an associated SOA module.

1. Introduction

Volatile organic compounds (VOCs) are emitted from both anthropogenic and biogenic sources (e.g. Guenther et al., 1995; Dore et al., 2003) and have a major influence on the chemistry of the lower atmosphere. It is well documented that the atmospheric degradation of VOCs plays a central role in the generation of ozone, O_3 (e.g., Leighton, 1961; Finlayson-Pitts and Pitts, 2000; Atkinson, 2000; Jenkin and Clemitshaw, 2000), and leads to the formation of other secondary pollutants, most notably fine particulate matter in the form of secondary organic aerosol (SOA) (e.g., Kanakidou et al., 2005 and references therein). Ozone and fine particles have harmful impacts on human health and

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on the environment, and influence the Earth's radiation budget: consequently, a sound understanding of the chemistry of VOC degradation, and its adequate representation in atmospheric models, is of central importance to the development of environmental policy on local, regional and global-scale issues.

Emissions speciation data indicate that many hundreds of VOCs are emitted (e.g. Dore et al., 2003), which possess a variety of physico-chemical properties, by virtue of differences in structure and functional group content. Since these factors influence the reactivity (i.e., the rate of oxidation in the atmosphere) and the oxidation pathways available (i.e., the degradation mechanism), it has long been recognised that the propensities of VOCs to form secondary pollutants such as ozone and SOA vary from one compound to another (e.g., Derwent and Jenkin, 1991; Carter, 1994; Grosjean and Seinfeld, 1989). Modelling studies using detailed emitted VOC speciations and comprehensive descriptions of VOC degradation chemistry have thus provided a means of quantifying the roles played by VOCs (both individually and collectively) in atmospheric chemistry, thereby allowing detailed appraisals of the contributions made by individual VOCs, or VOC classes, to ozone and SOA formation (e.g., Johnson et al., 2006a, b; Derwent et al., 2007a, b). Those studies made use of the Master Chemical Mechanism, version 3.1 (MCM v3.1), a comprehensive chemical mechanism that describes the detailed degradation of more than 100 emitted VOCs (Jenkin et al., 1997, 2003; Saunders et al., 2003; Bloss et al., 2005). Because of its explicit nature, the MCM represents a direct method of utilising and applying the results of studies of elementary chemical processes, and is conceptually simple, since it contains almost no empirical lumping or surrogate species. As a result, however, the mechanism contains many thousands of chemical species and reactions, and it is recognised that the development of less-detailed schemes is essential for many applications where greater computational efficiency is required.

The development of a reduced mechanism to describe the formation of ozone from the oxidation of a similar set of emitted VOCs has previously been described by Jenkin et al. (2002a), the mechanism being known as the Common Representative Intermediates (CRI) mechanism. The construction methodology, which is outlined below, yielded a mechanism of ca. 250 species and 570 reactions (hereafter referred to as CRI v1), with the performance of the whole mechanism (in terms of ozone formation) being optimized in relation to that of the version of the MCM which was current at the time (MCM v2). A recent detailed appraisal of the performance of CRI v1 in relation to that of the latest MCM version (MCM v3.1), has shown that, although CRI v1 generally performs well, it tends to display under-reactivity under highly polluted conditions characteristic of close to emissions sources (Watson, 2007). In addition, the formation of ozone specifically from the oxidation of aromatic hydrocarbons at short timescales is also severely underestimated in comparison with MCM v3.1, improvement of that aspect of the chemistry being one of the major developments in the more recent versions of the MCM (Jenkin et al., 2003; Bloss et al., 2005).

In the present paper, and the companion papers (Watson et al., 2008; Utembe et al., in preparation), the development and application of version 2 of the CRI mechanism (hereafter referred to as CRI v2) is described. The present paper describes the development of the gas phase CRI v2 mechanism itself, and optimisation of its performance in relation to MCM v3.1, to yield a mechanism of intermediate complexity which can degrade methane and 115 emitted non-methane hydrocarbons and oxygenated VOCs. A series of emissions lumping options are considered in Watson et al. (2008), yielding a traceable set of further reduced CRI mechanisms, the smallest of which is considered appropriate for application as a reference mechanism in global chemistry-transport models. Finally, the development and assessment of an associated SOA module, for application with CRI v2 and its reduced variants, is described in Utembe et al. (in preparation).

2. Quantifying ozone formation from VOC oxidation: the CRI index

The complete oxidation of a given VOC through to CO_2 and H_2O normally proceeds via a series of intermediate oxidised products. At each stage, the chemistry can be propagated by reactions of free radicals leading to the oxidation of NO to NO_2 and resultant formation of O_3 as a by-product, following the photolysis of NO_2:

$$NO_2 + h\nu \;\; (\lambda < 420 \;\; \mathrm{nm}) \rightarrow NO + O(^3P) \tag{1}$$

$$O(^3P) + O_2(+M) \rightarrow O_3(+M) \tag{2}$$

The total quantity of O_3 potentially generated from the free radical propagated chemistry is therefore dependent on the number of NO-to-NO_2 conversions which can occur during the degradation. As described previously (Jenkin et al., 2002a), understanding of the detailed chemistry for simple hydrocarbons suggests that the total molecular yield of O_3 potentially formed as a by-product during the complete OH-initiated and NO_x-catalysed oxidation of a molecule of the hydrocarbon through to CO_2 and H_2O is equivalent to the number of reactive bonds in the parent molecule (i.e., the number of C–H and C–C bonds which are eventually broken during the complete oxidation to CO_2 and H_2O). Accordingly, in the simplest case of methane, the overall chemistry can be represented by the following equation:

$$CH_4 + 8O_2 \rightarrow CO_2 + 2H_2O + 4O_3 \tag{3}$$

yielding four molecules of O_3, which is based on the sequential oxidation via HCHO and CO as intermediate oxidised products, with conversion of NO to NO_2 resulting from reactions of the peroxy radicals (CH_3O_2 and HO_2) with NO, as shown in Fig. 1. In a similar way, Jenkin et al. (2002a) demonstrated that degradation of ethane can be represented by the following overall chemistry, yielding seven molecules of O_3:

$$C_2H_6 + 14O_2 \rightarrow 2CO_2 + 3H_2O + 7O_3 \tag{4}$$

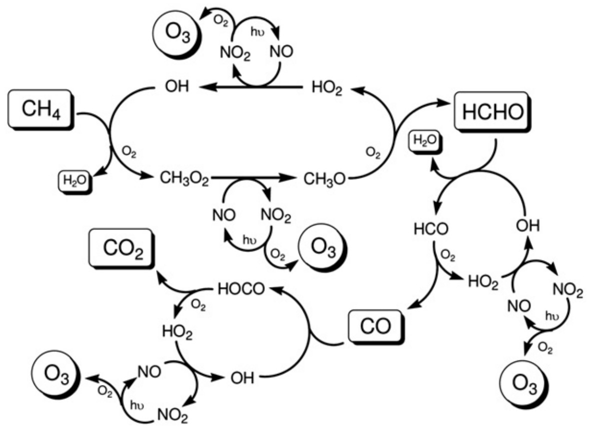

Fig. 1. Schematic representation of the reactions involved in the OH initiated, NO_x catalysed oxidation of methane to CO_2 and H_2O. The reaction sequence shows the formation of formaldehyde (HCHO) and CO as intermediate oxidised products. The complete sequence results in the formation of four ozone molecules, with the net overall chemistry as shown in reaction (3). (NB: the oxidation of CO to CO_2 partially proceeds via $OH + CO \rightarrow H + CO_2$, followed by $H + O_2 + M \rightarrow HO_2 + M$, which has the same overall chemistry as that displayed above, i.e. $OH + CO + M \rightarrow HOCO + M$, followed by $HOCO + O_2 \rightarrow HO_2 + CO_2$).

On this basis, the number of reactive (C—H and C—C) bonds in a closed-shell intermediate can be used as an index to represent the number of NO-to-NO_2 conversions which can potentially result from its subsequent OH-initiated and NO_x-catalysed degradation. In the CRI mechanism, this simple rule is used to define a series of generic intermediates, which can mediate the breakdown of larger VOCs (Jenkin et al., 2002a). A single intermediate can therefore be used as a "common representative" for a large set of species possessing the same index, as formed in detailed mechanisms such as the MCM. The primary size saving in the CRI mechanism therefore results from simplification of the highly detailed chemistry for larger and more complex VOCs. As described in detail by Jenkin et al. (2002a), CRI v1 was constructed on this basis, and was subsequently tested in its entirety by comparison with MCM v2, using simulations carried out with a Photochemical Trajectory Model (PTM) operating on a standard trajectory over north-west Europe. The agreement was optimised by varying the rate coefficients for the reactions of OH with the limited number of generic intermediate carbonyl compounds in CRI v1, using total ozone formation as the primary criterion of performance.

The construction of CRI v2 is outlined in the following sections. In contrast to the procedure adopted for CRI v1, CRI v2 was built up on a compound-by-compound basis, with the performance of the chemistry optimised for each compound in turn by comparison with that of MCM v3.1. This more stringent examination in relation to the detailed chemistry revealed that there are certain instances when the simple rule outlined above is not followed; namely when an intermediate peroxy radical undergoes a rearrangement. In MCM v3.1, this occurs for vinyl-type radicals (formed, for example, during the degradation of dialkenes such as isoprene and 1,3-butadiene) and for hydroxycyclohexadienyl peroxy radicals formed from the addition of OH to aromatics (Saunders et al., 2003; Jenkin et al., 2003). Because this chemistry bypasses an NO-to-NO_2 conversion which could otherwise occur, and results in loss

Table 1
One hundred and fifteen emitted non-methane hydrocarbons and oxygenated VOCs degraded in CRI v2. Biogenic emissions are represented by isoprene, α-pinene and β-pinene, with the remaining species representing anthropogenic emissions

Alkanes (21)	Alkenes/alkynes (20)	Aromatics (18)	Oxygenates (56)	
Ethane	Ethene	Benzene	Formaldehyde	2-Methyl-2-butanol
Propane	Propene	Toluene	Acetaldehyde	Cyclohexanol
Butane	1-Butene	Ethylbenzene	Propanal	4-Hydroxy-4-methyl-2-pentanone
Pentane	1-Pentene	Propylbenzene	Butanal	Ethane-1,2-diol
Hexane	1-Hexene	*i*-Propylbenzene	Pentanal	Propane-1,2-diol
Heptane	*cis*-2-Butene	*o*-Xylene	Methylpropanal	Dimethylether
Octane	*trans*-2-Butene	*m*-Xylene	Propanone	Diethylether
Nonane	*cis*-2-Pentene	*p*-Xylene	Butanone	Methyl-*t*-butylether
Decane	*trans*-2-Pentene	1,2,3-Trimethylbenzene	2-Pentanone	Di-*i*-propylether
Undecane	*cis*-2-Hexene	1,2,4-Trimethylbenzene	3-Pentanone	Ethyl-*t*-butylether
Dodecane	*trans*-2-Hexene	1,3,5-Trimethylbenzene	2-Hexanone	2-Methoxyethanol
Methylpropane	Methylpropene	*o*-Ethyltoluene	3-Hexanone	2-Ethoxyethanol
Methylbutane	2-Methyl-1butene	*m*-Ethyltoluene	3-Methyl-2-butanone	1-Methoxy-2-propanol
2,2-Dimethylpropane	3-Methyl-1-butene	*p*-Ethyltoluene	4-Methyl-2-pentanone	2-Butoxyethanol
2-Methylpentane	2-Methyl-2-butene	3,5-Dimethylethylbenzene	3,3-Dimethyl-2-butanone	1-Butoxy-2-propanol
3-Methylpentane	1,3-Butadiene	3,5-Diethyltoluene	Cyclohexanone	Dimethoxymethane
2,2-Dimethylbutane	Isoprene	Benzaldehyde	Methanol	Dimethylcarbonate
2,3-Dimethylbutane	α-Pinene	Styrene	Ethanol	Formic acid
2-Methylhexane	β-Pinene		1-Propanol	Acetic acid
3-Methylhexane	Ethyne		2-Propanol	Propanoic acid
Cyclohexane			1-Butanol	Methylformate
			2-Butanol	Methylacetate
			2-Methyl-1-propanol	Ethylacetate
			2-Methyl-2-propanol	*n*-Propylacetate
			3-Pentanol	*i*-Propylacetate
			3-Methyl-1-butanol	*n*-Butylacetate
			3-Methyl-2-butanol	*s*-Butylacetate
			2-Methyl-1-butanol	*t*-Butylacetate

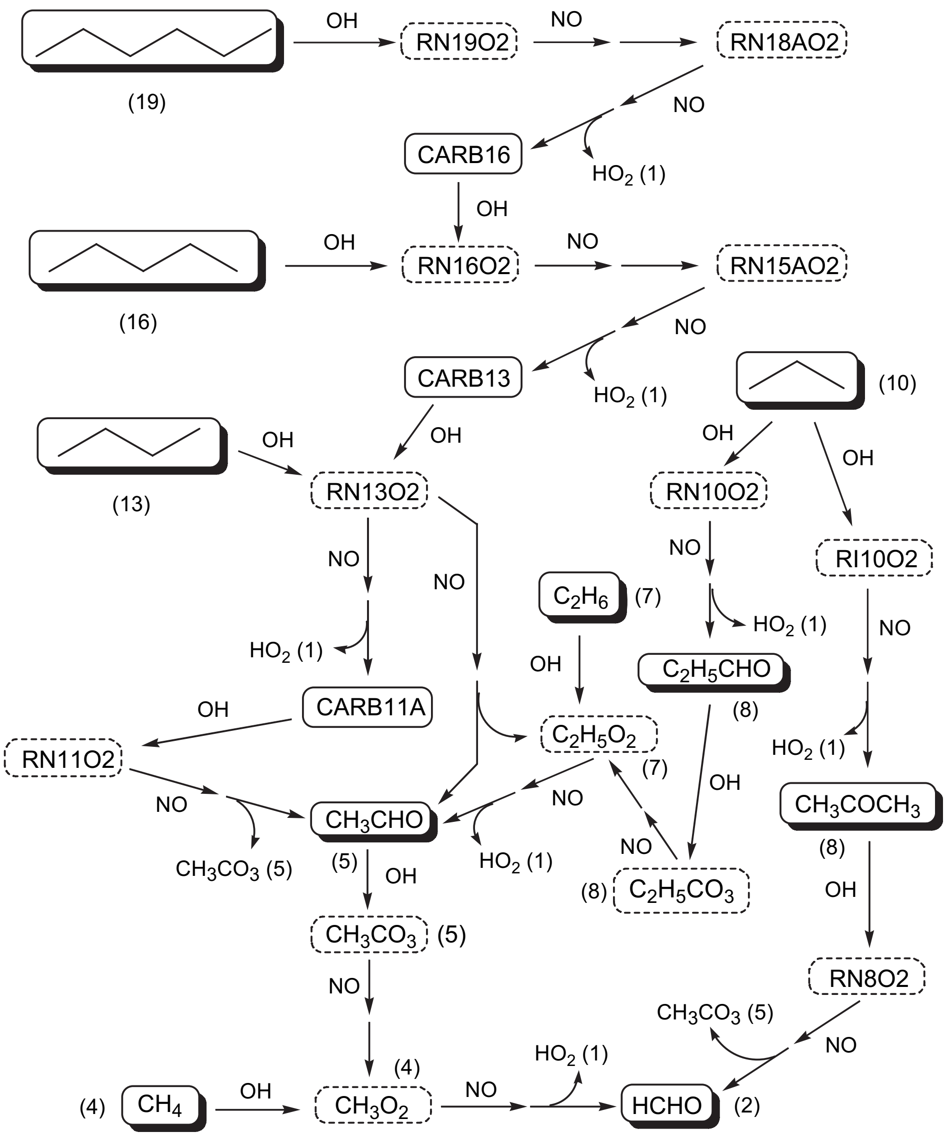

Fig. 2. Schematic representation of the OH-initiated, NO_x catalysed degradation chemistry for C_1–C_6 alkanes in CRI v2 (see discussion in text).

of a reactive bond, it reduces the CRI index by two. The sequential construction–optimisation procedure adopted for CRI v2 therefore allowed class-specific features such as these to be incorporated, and ensured that the contributions of individual VOCs, and VOC classes, to secondary pollutant formation were well represented.

3. Construction and testing of the CRI mechanism, version 2

3.1. Overview of construction procedure

CRI v2 was built up on a compound-by-compound basis, with its performance optimised to that of MCM v3.1, using a series of five-day box model simulations. The model represents a well-mixed boundary layer box, 1 km in depth, which receives emissions of NO_x, CO, SO_2, methane and non-methane VOCs, based on average emission densities in the UK in 2001 (Dore et al., 2003), but with appropriate diurnal and seasonal variations superimposed, as presented by Utembe et al. (2005). The model was run for conditions appropriate to a latitude of 51.5°N on June 21, under clear sky conditions. The reference simulations were obtained by running a version of the model containing MCM v3.1, with the non-methane VOCs being emitted entirely in the form of each compound in turn, for the 115 hydrocarbons and oxygenated VOCs listed in Table 1. Removal of ozone by dry deposition was also represented, leading to an ozone lifetime of ca. 2.5 days with respect to this process. All simulations in the present study were carried out using the FACSIMILE for Windows kinetics integration package, version 3.5 (MCPA Software).

The CRI v2 schemes were constructed for each compound in turn, starting with the smallest compound in each class, with the performance for each VOC optimised to that of MCM v3.1 using non-linear least squares fitting. In each case, the OH reactivity and photolysis rate of key intermediates were varied to optimise the agreement between the performance of the CRI v2 and MCM v3.1 schemes, with formation of ozone used as the primary criterion. The optimised parameters for the given intermediates were then fixed, prior to moving onto the next (larger) compound. The degradation scheme for the larger compound generates new intermediates, for which parameters can be optimised, but invariably feeds into the chemistry predefined for smaller compounds. The objective of this procedure, therefore, was to build up a representation of the degradation of the suite of emitted VOCs, whilst aiming to minimise the number of intermediates that needed to be defined to allow an acceptable quantification of ozone formation in comparison with MCM v3.1. The resultant CRI v2 mechanism contains 434 species and 1183 reactions (compared with 4,361 species and 12,775 reactions to degrade the same set of VOCs in MCM v3.1). This is approaching a factor of two larger than CRI v1 (ca. 250 species and 570 reactions), which reflects the implementation of a larger number of class-specific sets of intermediates in CRI v2.

As an illustration, some of the degradation chemistry for C_1–C_6 *n*-alkanes is presented schematically in Fig. 2, showing the major NO_x-catalysed oxidation pathways following the attack of OH. The index assigned (in brackets) to each alkane is the total number of C—H and C—C bonds it contains. The indices within the generic radicals and carbonyls (e.g. '19' in *RN19O2* and '16' in *CARB16*), thus represent the number of NO to NO_2 conversions which can result from the subsequent complete degradation via OH-initiated, NO_x-catalysed chemistry. As a result, the summed indices of the products of the reaction of a given RO_2 with NO is one less than the index of the RO_2 radical itself, because one NO-to-NO_2 conversion has occurred. HO_2 is assigned an index of 1 in the figure, since a single NO-to-NO_2 conversion regenerates OH, which initiated the reaction sequence. As indicated in the previous section, the CRI approach allows substantial simplification of the highly detailed chemistry for larger and more complex VOCs. However, the chemistry of smaller compounds (e.g., methane, ethane and propane in Fig. 2) is essentially identical to that in MCM v3.1.

Fig. 2 shows only the major radical-propagated oxidation pathways, initiated by reaction with OH at each stage, and propagated by the reactions of RO_2 radicals with NO. Other associated reactions can also propagate the chemistry, or lead to net radical removal or production, and these also need to be included if the impact of VOC oxidation is to be adequately represented. Fig. 3 shows a more complete representation of the CRI v2 chemistry for the example of pentane, and how it feeds into the chemistry for butane (which was pre-defined and optimised during the construction procedure). This shows that propagation reactions of the RO_2 radicals with NO_3 and with the peroxy radical "pool" are also represented, and that competing termination reactions are included for the RO_2 radicals with NO (leading to the formation of nitrates, $RONO_2$) and with HO_2 (leading to the formation of hydroperoxides). As discussed in the MCM construction protocol (Jenkin et al., 1997; Saunders et al., 2003), the branching ratios or rate coefficients for these reactions depend on the size of the RO_2 radical. In CRI v2, the assigned size-dependent coefficient was based on the weighted average for the alkyl peroxy radicals with the equivalent index. The nitrate and hydroperoxide species act as temporary reservoirs for free radicals, being degraded by photolysis and by reaction with OH. Once again, products of these reaction classes possessing appropriate CRI indices were generally defined on the basis of the chemistry in MCM v3.1, and weighted

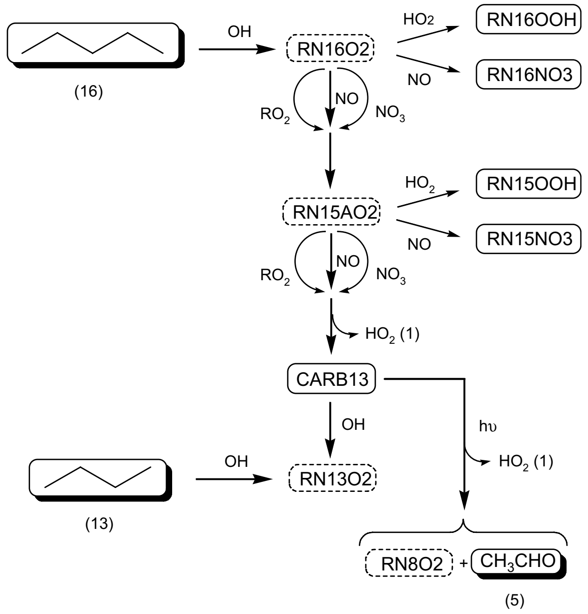

Fig. 3. Schematic representation of the initial stages of the degradation chemistry developed for pentane as part of the CRI v2 construction procedure. This demonstrates how the free-radical propagated chemistry feeds into the pre-defined chemistry for the smaller alkane, butane; and also displays selected competing chain termination reactions for the intermediate peroxy radicals (RN16O2 and RN15AO2).

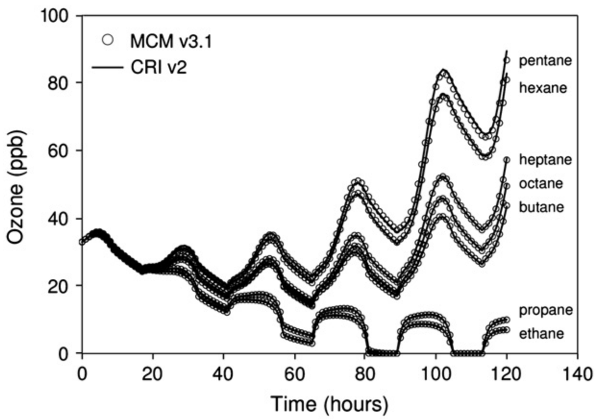

Fig. 4. Ozone profiles simulated using MCM v3.1 and CRI v2, with non-methane VOCs emitted entirely in the form of each of the C_2–C_8 *n*-alkanes. The simulations were initialised with 33 ppb ozone and 2 ppb NO_x, and started at midday.

average rate parameters were applied to the reactions. A key intermediate in the pentane degradation chemistry in Fig. 3 is the carbonyl intermediate, *CARB13*. This is degraded either by reaction with OH to generate the peroxy radical *RN13O2* (thereby feeding directly into the pre-optimised butane chemistry), or by photolysis to yield the smaller pre-defined peroxy radical *RN8O2* (see Fig. 2), CH_3CHO and HO_2. Because the photolysis reaction is a radical source, the total CRI index of these latter products is increased by one unit relative to *CARB13*. Consequently, the simulated formation of ozone from pentane degradation is sensitive to the absolute and relative rates assigned to photolysis and OH reaction for *CARB13*. These parameters were varied simultaneously, using non-linear least squares fitting to optimise the agreement between the ozone profiles generated using CRI v2 and MCM v3.1, when non-methane VOC emissions were entirely in the form of pentane. Fig. 4 shows a comparison of the MCM v3.1 and optimised CRI v2 ozone profiles for all C_2–C_8 *n*-alkanes, which demonstrates the good level of agreement that was obtained using this procedure.

Similar procedures were thus used to construct the degradation chemistry for all 115 hydrocarbons and oxygenated VOCs listed in Table 1, with a representation of ozone and NO_3-initiated degradation also included where appropriate (i.e., primarily for alkenes and dialkenes). Fig. 5 shows a comparison of the average ozone mixing ratios simulated with the optimised CRI v2 and MCM v3.1 for all 115 hydrocarbons and oxygenated VOCs listed in Table 1. This demonstrates that acceptable agreement was achieved for the wide variety of component VOCs. In some cases, the slightly poorer agreement is indicative of a level of compromise in CRI v2, where significant improvement would require a substantial expansion of the chemistry. In particular, it proved difficult to construct substantially reduced chemistry for the monoalkyl-substituted benzenes (e.g., toluene), which could recreate the MCM v3.1 ozone profiles over a five-day period with the sustained precursor input. This reflects the complexity of the chemistry of the aromatics in general, which is most pronounced for the monoalkyl-substituted benzenes because several of the identified aromatic degradation pathways make significant

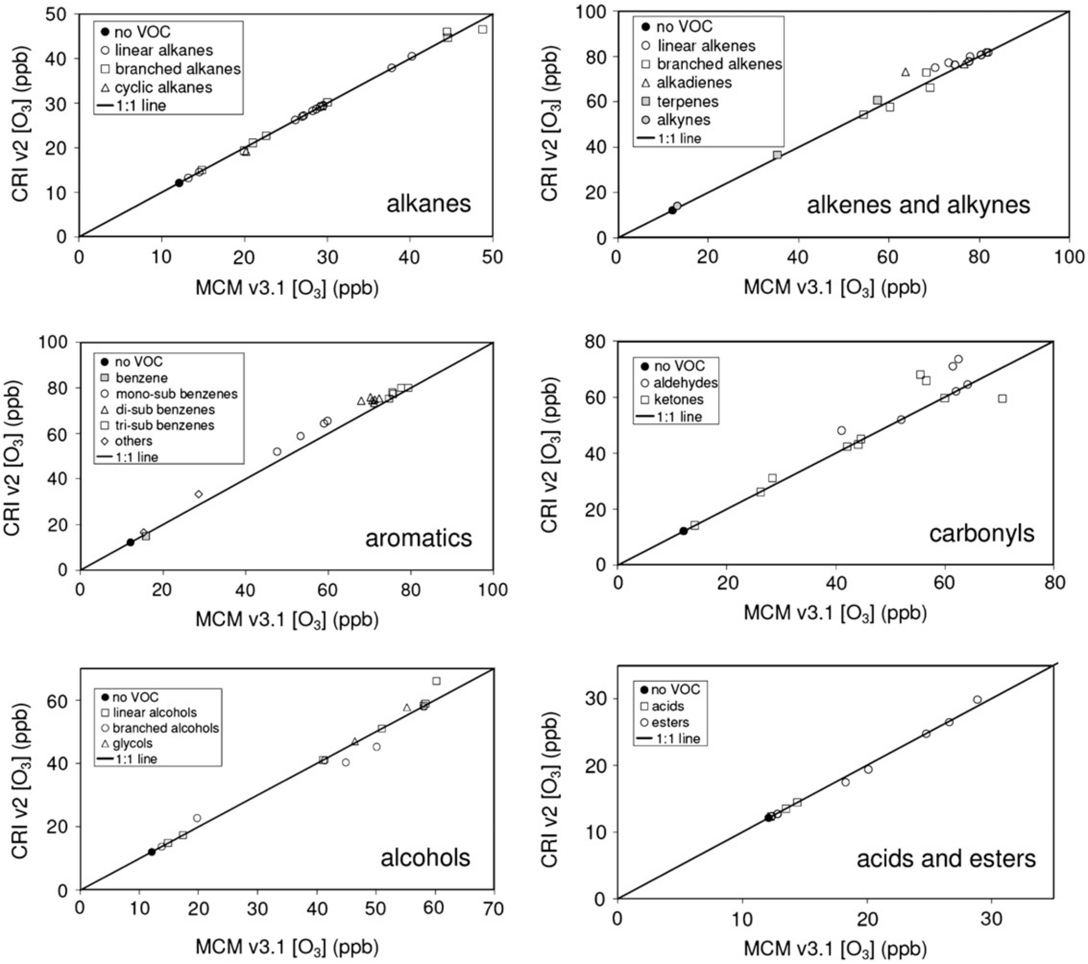

Fig. 5. Comparison of average ozone mixing ratios simulated using MCM v3.1 and CRI v2, with non-methane VOCs emitted entirely in the form of each of the 115 hydrocarbons and oxygenated VOCs listed in Table 1. The average ozone mixing ratio when non-methane VOC emissions were omitted is also displayed in each panel. All simulations were initialised with 33 ppb ozone and 2 ppb NO_x.

contributions for these species, leading to notable formation of both ring-opened and ring-retained products as represented in MCM v3.1 (Bloss et al., 2005). In these cases, therefore, the performance of the reduced chemistry was optimised to give good agreement for the first two days of the simulation, with the schemes generally overestimating the ozone mixing ratios subsequently. Given the general uncertainties in the understanding and representation of aromatic degradation chemistry in chemical mechanisms, and their ability to simulate ozone formation, this is regarded as an acceptable compromise for the current state of knowledge.

3.2. Testing of anthropogenic VOC chemistry

Fig. 6 shows a comparison of ozone profiles simulated with CRI v2 and MCM v3.1 throughout the five-day period for the complete speciation of 112 non-methane anthropogenic VOCs shown in Table 1. The relative contributions of the individual VOCs to the emissions total were based on the UK NAEI speciation, as reported by Utembe et al. (2005), but with the comparatively small collective contribution (ca. 4%) from a limited number of chlorinated hydrocarbons omitted, because CRI v2 has not been developed to treat this class of compound. Given the generally low propensities of the chlorinated compounds to generate ozone (e.g., Derwent et al., 2007b), and their small collective contribution to the VOC emissions total, their omission has a very minor influence on the simulated ozone formation.

Simulations were carried out for the standard emissions densities, and with the VOC/NO_x emission ratio varied over a range of 32 by scaling the NO_x emissions by factors between 0.125 and 4. This allowed the transition from VOC-limited to NO_x-limited conditions to be examined, which is presented in terms of the VOC/NO_x ratio relative to the base case, $(VOC/NO_x)_{rel}$, in Fig. 6. The ozone profiles simulated with CRI v2 are in good agreement with the MCM v3.1 profiles for the complete range of VOC/NO_x emission ratio. The base case profile shows a tendency to overestimation (by ca. 5%) in the latter stages of the simulation, mainly due to the compromise outlined above for the aromatic chemistry. In all the other scenarios the level of agreement is better than in the base case.

As described in detail elsewhere (Watson, 2007), wider testing of the performance of CRI v2 has been carried out in relation to simulations of ozone, NO, NO_2, OH, HO_2 RO_2, NO_3, HNO_3, H_2O_2 and HCHO. This has been done for the complete anthropogenic VOC speciation, and also for VOC emissions entirely as alkanes, alkenes, aromatics and oxygenates, whilst maintaining the relative speciation within these categories in line with the UK NAEI. In each case, variation of the VOC/NO_x emissions ratio over at least a range of four has been considered. This has demonstrated a generally good performance over a wide range of conditions, with the simulations of ozone typically agreeing to better than 10% for the simulations in which VOC emissions were emitted entirely as aromatics, and to better than 5% in all other cases. The example comparisons shown in Figs. 7 and 8a demonstrate that the simulated levels of OH and HCHO for the full anthropogenic VOC mix also typically agree to better than 5% for the complete range of VOC/NO_x emissions ratio discussed above. In all respects, CRI v2 was found to perform better than CRI v1 in comparison with MCM v3.1, with particular improvements in performance

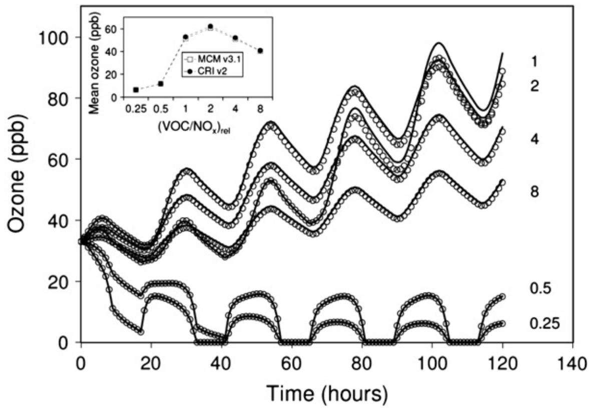

Fig. 6. Comparison of ozone mixing ratios simulated from anthropogenic VOC degradation over a five-day period with MCM v3.1 (○) and CRI v2 (solid line) for a range of 32 in VOC/NO_x emission ratio (June 21 conditions, 51.5°Lat). The base case simulation was based on average 2001 UK emission densities for anthropogenic VOCs and NO_x, using the NAEI VOC speciation (see text), the applied daily average densities being 15.4 kg km^{-2} day^{-1} for VOCs and 18.3 kg km^{-2} day^{-1} for NO_x. The range in VOC/NO_x was achieved by scaling the NO_x emissions, and is displayed relative to the base case, denoted $(VOC/NO_x)_{rel}$. The inset shows a comparison of the simulated mean ozone mixing ratios.

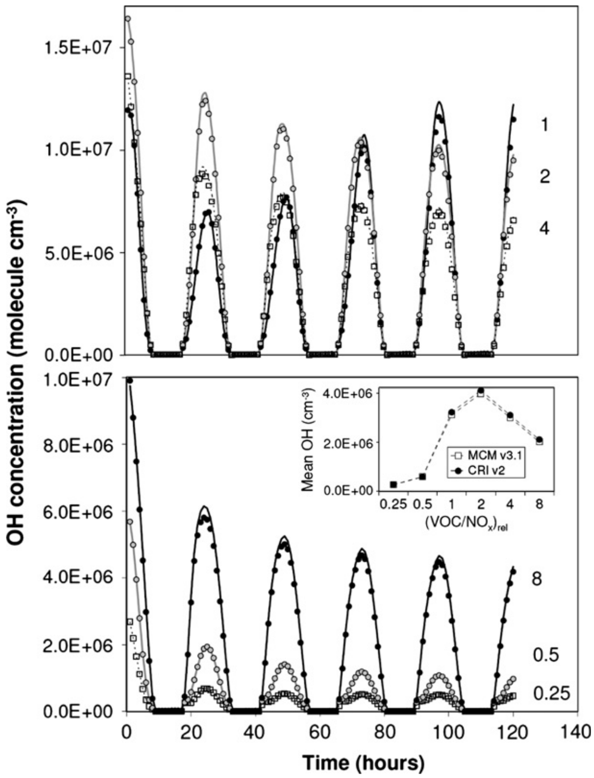

Fig. 7. Comparison of OH concentrations simulated over a five-day period with MCM v3.1 (points) and CRI v2 (lines), for the conditions summarised in the Fig. 6 caption. The lower panel inset shows a comparison of the simulated mean OH concentrations.

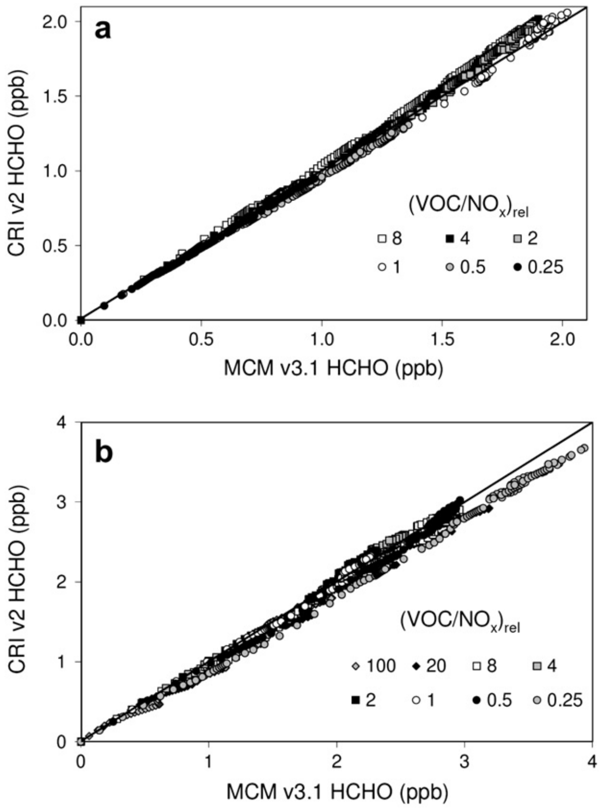

Fig. 8. Correlation of HCHO mixing ratios generated from the processing of emitted VOCs over a five-day period, simulated with MCM v3.1 and CRI v2. (a) Results using an anthropogenic VOC speciation, for the conditions summarised in the Fig. 6 caption; (b) Results using a biogenic VOC speciation, for the conditions summarised in the Fig. 10 caption. The 1:1 line in also shown in each panel.

under highly VOC-limited conditions (i.e., elevated NO_x emissions) and when non-methane VOC emissions were entirely in the form of aromatics (see Watson, 2007).

3.3. Testing of biogenic VOC chemistry

Fig. 9 shows a comparison of base case ozone profiles simulated with CRI v2 and MCM v3.1 for non-methane VOC emissions entirely in the form of each of the three biogenic hydrocarbons, isoprene, α-pinene and β-pinene. This shows excellent agreement in the case of isoprene, but a degree of compromise for the pinenes. Similarly to the aromatics discussed above, it proved difficult to construct substantially reduced chemistry for these species which could recreate the MCM v3.1 ozone profiles over a five-day period with the sustained precursor input, reflecting the notable complexity of the chemistry. Once again, it was possible for the performance of the reduced CRI v2 chemistry to be optimised to give good agreement for the first two or three days of the simulation, with the schemes overestimating the ozone mixing ratios subsequently. Despite this, the agreement is still considered to be good, given the general uncertainties in the understanding and representation of α- and β-pinene degradation chemistry in chemical mechanisms (e.g., Pinho et al., 2007).

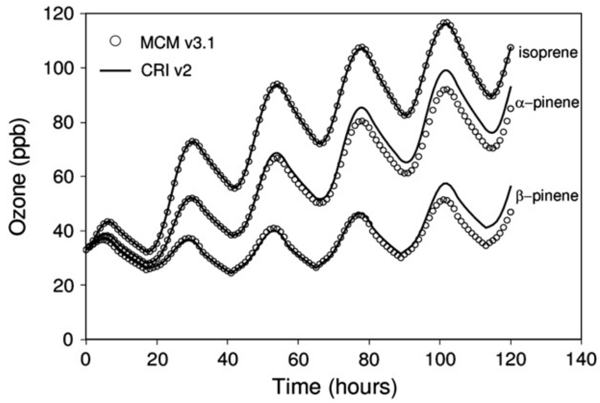

Fig. 9. Ozone profiles simulated using MCM v3.1 and CRI v2, with non-methane VOCs emitted entirely in the form of each of isoprene, α-pinene and β-pinene.

Fig. 10 shows a comparison of ozone profiles simulated with CRI v2 and MCM v3.1 for a range of VOC/NO_x emissions ratios, using a previously applied biogenic hydrocarbon speciation for isoprene and monoterpenes (Utembe et al., 2005). For procedural consistency, the applied VOC emissions density was based on the UK anthropogenic total. Simulations were carried out for a substantially extended range of VOC/NO_x emission ratios of 400 (obtained by scaling the NO_x emissions by factors between 0.01 and 4), thus allowing the transition from VOC-limited through to highly NO_x-limited conditions to be examined. As shown in Fig. 10, the ozone profiles simulated with CRI v2 are once again in acceptable agreement with the MCM v3.1 profiles for the complete range of VOC/NO_x emission ratio. Importantly, the abilities of the mechanisms to sequester NO_x in the form of higher oxidised (inorganic and organic) nitrogen compounds was found to be comparable across the VOC/NO_x range,

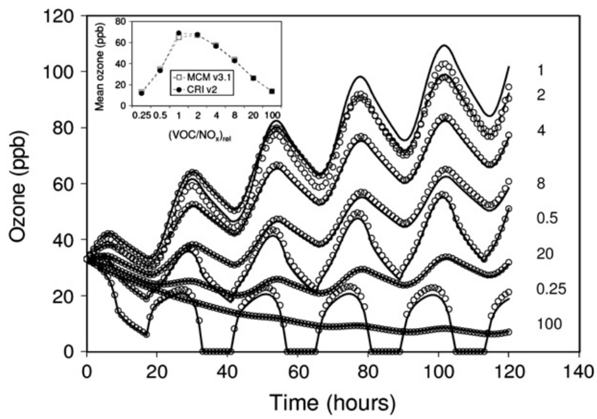

Fig. 10. Comparison of ozone mixing ratios simulated from biogenic VOC degradation over a five-day period with MCM v3.1 (○) and CRI v2 (solid line) for a range of 400 in VOC/NO_x emission ratio (June 21 conditions, 51.5°Lat). VOCs were emitted as 50% isoprene, 30% α-pinene and 20% β-pinene. The range in VOC/NO_x was achieved by scaling the NO_x emissions, and is displayed relative to the base case, denoted $(VOC/NO_x)_{rel}$. The inset shows a comparison of the simulated mean ozone mixing ratios.

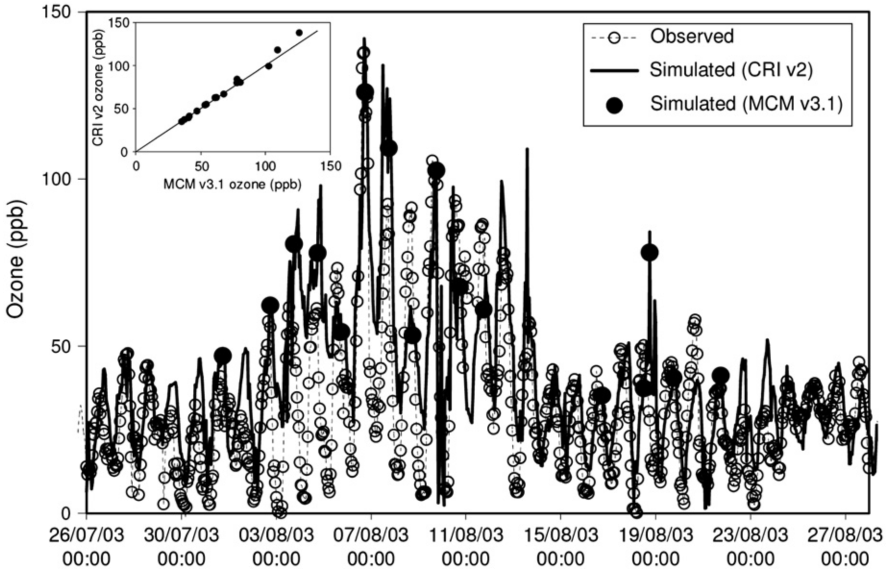

Fig. 11. Comparison of observed hourly mean ozone mixing ratios with those simulated for the entire TORCH campaign with the PTM containing CRI v2, and for 16 case study trajectories with the PTM containing MCM v3.1. The ozone mixing ratios were measured by the University of York, as reported in Utembe et al. (2005).

confirming that the biogenic mechanisms would have similar impacts on ozone chemistry under the highly NO_x-limited conditions characteristic of more remote locations where significant biogenic inputs occur. The formation of HCHO was also found to be in good agreement for the full VOC/NO_x range (see Fig. 8b), indicating that CRI v2 would provide an acceptable basis for interpretation of space-based HCHO measurements in relation to constraining VOC emissions, and their impacts, in remote and polluted environments (e.g., Fu et al., 2007; Palmer et al., 2007).

3.4. Performance under simulated ambient conditions

The CRI v2 mechanism was finally tested to examine its ability to simulate ozone formation in comparison with MCM v3.1, in relation to observations made during the Tropospheric Organic Chemistry Experiment (TORCH) campaign at Writtle, Essex, UK (51.74°N; 0.42°E), a rural site approximately 40 km to the north-east of central London, during late July and August 2003 (Lee et al., 2006). This was done using a Photochemical Trajectory Model (PTM) which has been widely applied to the simulation of photochemical ozone formation in north-west Europe (e.g., Derwent et al., 1998; Jenkin et al., 2002b), and which has recently been used simulate the TORCH 2003 campaign in a number of studies (Utembe et al., 2005; Johnson et al., 2006a, b; Jenkin et al., 2008). The PTM containing CRI v2 was used to simulate the chemical development of boundary layer air parcels being advected along 96-hour trajectories arriving at the Writtle site at hourly resolution for the entire campaign period. This made use of ca. 800 back-trajectories, which were obtained from the NOAA on-line trajectory service (Draxler and Rolph, 2003). An otherwise identical version of the PTM containing MCM v3.1 was also used to simulate 16 case study events during the campaign period.

The simulated ozone mixing ratios at the receptor are presented in Fig. 11, along with the hourly-mean observations. The CRI v2 version of the model recreates the general features of the observed time series, including the elevated ozone levels which prevailed during a stable anticyclonic period of the campaign (3–12 August), indicating that the model is able to provide a reasonable representation of the conditions experienced. As indicated in the figure, the simulated ozone mixing ratios agree well with those simulated using the MCM v3.1 version of the model for the 16 case study events. A correlation of the simulated data (inset to Fig. 11) emphasises this good agreement, and demonstrates a slight tendency towards overestimation in the highest ozone events. These events correspond to slow-moving trajectories which have circulated over north-west Europe during the anticyclone, and which have therefore received more sustained precursor input over the period of the back trajectory. Consequently, this slight overestimation is consistent with that discussed above for the latter stages of the box model simulations. The 16 selected case studies represent a variety of conditions, such that the inputs of VOCs and NO_x each vary over a range of more than an order of magnitude, with the VOC/NO_x ratios varying from ca. 0.8 to ca. 2, and with the fractional contribution of biogenic VOC to the VOC total varying from ca. 5% to ca. 30%.

4. Summary and conclusions

A reduced mechanism describing ozone formation from the tropospheric degradation of methane and 115 emitted non-methane hydrocarbons and oxygenated VOCs has been developed, using MCM v3.1 as a reference benchmark. The CRI v2 mechanism was built up on a compound-by-compound basis, with the performance of its chemistry optimised for each compound in turn by comparison with MCM v3.1, using formation of ozone as the primary criterion. The resultant mechanism contains 1183 reactions of 434 chemical species, and is therefore about an order of magnitude smaller than MCM v3.1. As described in Section 3, the CRI v2 construction procedure allowed the ozone forming abilities of the wide variety of component VOCs to be well described, with the complete mechanism providing box model simulations of the formation of ozone (and associated species) which are in acceptable agreement with MCM v3.1 over a five-day period for average UK conditions.

The performance of CRI v2 also compares well with that of MCM v3.1 over a wide range of ambient conditions in box model simulations. Variation of the VOC/NO_x emission ratio over a range of 32, in conjunction with a detailed anthropogenic VOC speciation, demonstrated that the mechanisms displayed an almost identical transition from VOC-limited to NO_x-limited conditions. Similar simulations using a biogenic VOC speciation, with the VOC/NO_x emission ratio varied over a range of 400, also demonstrated that the transition from VOC-limited through to highly NO_x-limited conditions was comparable and confirmed that the biogenic mechanisms would have similar impacts on ozone chemistry under the highly NO_x-limited conditions characteristic of more remote locations where significant biogenic inputs occur. The good agreement in the performance of the two mechanisms was further demonstrated in simulations of the TORCH 2003 campaign using a photochemical trajectory model, which considered notable ranges in the relative and absolute emissions of NO_x and VOCs, and in the relative contributions of anthropogenic and biogenic species to the VOC total.

CRI v2 is thus regarded as a reduced mechanism of intermediate complexity, which is traceable to MCM v3.1, and which provides a sound basis for further systematic reduction. Such reduction, on the basis of tested emissions lumping, is considered in the first of two companion papers (Watson et al., 2008), with the implementation of a secondary organic aerosol module described in the second paper (Utembe et al., in preparation).

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